

DIGITALIZED VOTER-ID SYSTEM WITH FINGERPRINT AUTHENTICATION AND INTRUSION DETECTION

A PROJECT REPORT

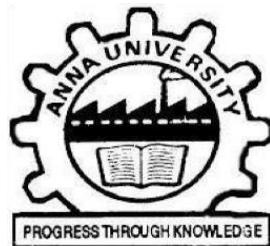
Submitted by

Vishwa R	2022505019
Jemmy Angel K J	2022505027
Vishal K	2022505055

*in partial fulfillment for the award of the degree
of*

**BACHELOR OF ENGINEERING
IN**

ELECTRONICS AND INSTRUMENTATION ENGINEERING



**DEPARTMENT OF INSTRUMENTATION ENGINEERING
MADRAS INSTITUTE OF TECHNOLOGY
ANNA UNIVERSITY: CHENNAI-600 044**

NOVEMBER 2025

MADRAS INSTITUTE OF TECHNOLOGY
ANNA UNIVERSITY: CHENNAI 600 044
BONAFIDE CERTIFICATE

Certified that this report titled “**DIGITALIZED VOTER-ID SYSTEM WITH FINGERPRINT AUTHENTICATION AND INTRUSION DETECTION**” is the Bonafide work of

Vishwa R **2022505019**

Jemmy Angel K J **2022505027**

Vishal K **2022505055**

who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr. S. SRINIVASAN,

HEAD OF THE DEPARTMENT,

Dept of Instrumentation Engineering,

Madras Institute of Technology Campus,

Anna University,

Chennai – 600 044.

SIGNATURE

Dr. S. NAMBI NARAYANAN,

SUPERVISOR,

Dept of Instrumentation Engineering,

Madras Institute of Technology Campus,

Anna University,

Chennai – 600 044.

ACKNOWLEDGEMENT

We are highly thankful to **Dr.P JAYASHREE**, Dean, Madras Institute of Technology, Chennai for his valuable and in depth experienced advises.

We would like to express our respectful thanks to **Dr. S. SRINIVASAN**, Professor and Head, Department of Instrumentation Engineering, Madras Institute of Technology, Anna University. We are highly obliged to our project guide **Dr. S. NAMBI NARAYANAN**, Teaching Fellow, Department of Instrumentation Engineering, for giving us motivation, valuable inputs and timely advice for the project.

We wish to extend our sincere thanks to the panel members **Dr.LATHA**, Professor, **Dr.C. SHANTHI**, Assistant Professor(Sl.Gr), **Dr.S. MEYYAPPAN**, Associate Professor, and **Dr.S.AROCKIYA SUKANYA** , Teaching Fellow for their valuable suggestions and encouragement during the reviews.

We also thank the faculty advisor project coordinator, **Dr. A. GANESH RAM** Assistant professor (Sl.Gr.) and **Dr. M. MYTHILY**, Assistant Professor and all the teaching and non-teaching staff members of the department of the Department of Instrumentation Engineering, for their support in all aspects.

Vishwa R	2022505019
Jemmy Angel K J	2022505027
Vishal K	2022505055

ABSTRACT

This project presents the design and implementation of a Digitalized Voter-ID System with Fingerprint Authentication and Intrusion Detection, aimed at enhancing the security and reliability of voter verification in electoral processes. The proposed system employs a digital voter identification card embedded with a QR code, which serves as the initial verification element for fetching voter information from an encrypted database. An encrypted database is utilized to store voter profiles, verified enrollment records, and biometric fingerprint templates, ensuring data confidentiality, integrity and availability. A dual-layer authentication mechanism is implemented by combining QR-code scanning with fingerprint recognition to accurately validate voter identity. The system provides real-time status updates through an LCD display with timestamp, enabling clear communication during the authentication process. In addition, an integrated intrusion detection feature continuously monitors for unauthorized access attempts, further strengthening system security. This project includes the development of a structured troubleshooting workflow to support error handling and operational continuity. Comprehensive testing is conducted, covering both successful and unsuccessful authentication scenarios, to evaluate system performance and reliability. The results confirm that the proposed solution enhances accuracy, reduces impersonation risks, and offers a secure and efficient framework for modern voter identification. The vote results are stored in a secured manner which includes AES encryption and blockchain for integrality. Any malpractices will be detected by the intrusion detection system and generates alarm. The votes are stored in the external memory along with the timestamp in hashed manner.

TABLE OF CONTENTS

CHAPTER. NO	TITLE	PAGE NO
	ABSTRACT	iv
	LIST OF FIGURES	viii
	LIST OF ABBREVIATIONS	ix
1.	1.1 INTRODUCTION	1
	1.2 PROBLEM STATEMENT	1
	1.2.1 ISSUES IN THE EXISTING SYSTEM	1
	1.2.2 NEED FOR PROPOSED SOLUTION	1
	1.2.3 PROBLEM DEFINITION	2
	1.3 OBJECTIVE	2
	1.4 THRUST AREA	3
	1.5 SOCIAL RELEVANCE	3
	1.6 CONCLUSION	4
2.	LITERATURE REVIEW	5
	2.1 INTRODUCTION	5
	2.2 LITERATURE SURVEY	5
	2.2.1 PROPOSED METHODOLOGY	5
	2.2.2 EXISTING METHODOLOGY	5
	2.3 CONCLUSION	7
3.	COMPONENTS	8
	3.1 INTRODUCTION	8
	3.2 MICROCONTROLLER – ESP-32	9
	3.2.1 ADVANTAGES OF USING ESP-32	9
	3.3 R307s FINGERPRINT SENSOR MODULE	9
	3.3.1 SENSOR SPECIFICATION	9

	3.3.2 WORKING OF R307S	11
	3.3.3 ADVANTAGES OF R307S	12
	3.4 DISPLAY (16x2 LCD with I2C MODULE)	13
	3.4.1 SPECIFICATION	13
	3.4.2 WORKING OF 16x2 LCD WITH I2C	14
	3.4.3 ADVANTAGES OF THE LCD 16x2	14
	3.5 REAL-TIME COLOCK MODULE- DS3231	15
	3.5.1 RTC- SPECIFICATIONS	16
	3.5.2 WORKING OF RTC MODULE	16
	3.5.3ADVANTAGES OF THE DS3231	17
	3.6 MicroSD Card along with MODULE	17
	3.6.1 SPECIFICATIONS	18
	3.6.2 ROLE OF MicroSD CARD	18
	3.7 CONCLUSION	18
4.	QR CODE–BASED VERIFICATION	19
	4.1 INTRODUCTION	19
	4.1.1 ROLE OF QR CODE	19
	4.1.2 ADVANTAGES OF USING QR	20
	4.1.3 IMPORTANCE IN VERIFICATION	20
	4.2 CONCLUSION	21
5.	WEB PAGE	22
	5.1 INTRODUCTION	22
	5.2 FRONT END DEVELOPMENT	22
	5.3 BACK-END DEVELOPMENT	23
	5.4 SPECIAL CONCESSION FOR PwD	25
	5.5 NOVELTY	27
	5.6 CONCLUSION	27

6.	RESULT	28
	6.1 INTRODUCTION	28
	6.2 VOTE STATUS	28
	6.3 HARDWARE SETUP	29
	6.4 VOTE RESULTS	30
	6.5 CONCLUSION	30
7.	SUMMARY AND CONCLUSION	31
	7.1 SUMMARY	31
	7.2 CONCLUSION	32
	7.3 FUTURE SCOPE	33
	REFERENCES	34

LIST OF FIGURES

FIGURE NO.	FIGURE TITLE	PAGE NO.
3.1	PINOUT DIAGRAM OF ESP32.	9
3.2	R307S FINGERPRINT SENSOR MODULE	10
3.3	DISPLAY (16x2 LCD with I2C module)	13
3.4	REAL-TIME CLOCK MODULE (DS3231)	15
3.5	MicroSD with MODULE	17
4.1	MODEL OF QR-VOTER ID	21
5.1	LOGIN PAGE OF E-VOTING WEBSITE	24
5.2	SIGNUP PAGE OF E-VOTING WEBSITE	25
5.3	PICTURE OF VOTER ID WITH MEDICAL CERTIFICATE	26
5.4	MEDICAL CERTIFICATE	26
6.1	HASHED VOTE	28
6.2	STORED BLOCKS OF VOTES IN THE MICROSD	28
6.3	INITIAL STEP FOR AUTHENTICATION	28
6.4	ELIGIBILITY CHECK FOR THE PERSON	29
6.5	NOT ELIGIBLE MESSAGE IS DISPLAYED	29
6.6	RESULT DECLARATION	30
6.7	CHECKING FOR BLOCKCHAIN	30

LIST OF ABBREVIATIONS

S.No	ABBREVIATION	EXPANSION
1.	ESP32	Espressif System Processor
2.	LCD	Liquid Crystal Display
3.	I2C/I ² C	Inter-Integrated Circuit
4.	RTC	Real Time Clock
5.	DS3231	Dallas Semiconductor
6.	QR	Quick Response
7.	FRR	False Rejection Ratio
8.	FAR	False Acceptance Ration
9.	RAM	Random Access Memory
10.	SPI	Serial Peripheral Interface
11.	UART	Universal Asynchronous Receiver Transmitter
12.	USB	Universal Serial Bus
13.	PwD	People with Disability
14.	AES	Advanced Encryption Standard
15.	GPIO	General Purpose Input Output
16.	TTL	Transistor Transistor Logic
17.	SDA	Serial Data Line
18.	SCL	Serial Clock Line
19.	LED	Light Emitting Diode

CHAPTER 1

1.1 INTRODUCTION

This chapter deals with the problem that we are facing with our existing system and the solution along with the objectives.

1.2. PROBLEM STATEMENT

Traditional voter identification systems rely on physical voter-ID cards and manual verification by election personnel. These methods are vulnerable to human error, slow processing, lack of real-time authentication, and false votes. As a result, they fail to provide a foolproof mechanism for ensuring that only legitimate and eligible voters participate in elections.

1.2.1 ISSUES IN THE EXISTING SYSTEM

The existing voter verification process suffers from several limitations:

Susceptibility to voter impersonation and fraudulent identity representation.

Lack of biometric authentication, leading to weak verification accuracy.

Absence of encrypted data storage, increasing the risk of data manipulation and breaches.

No integrated intrusion detection, allowing unauthorized system access to go unnoticed.

Inefficient workflow due to manual verification, resulting in delays and operational inconsistencies.

These weaknesses compromise the transparency, reliability, and security of the electoral process.

1.2.2 NEED FOR THE PROPOSED SOLUTION

To overcome these challenges, a secure, digitalized, and automated voter

verification mechanism is essential. The system must ensure:

- Accurate voter identification using QR-embedded digital voter IDs.
- Strong second-level verification through fingerprint authentication.
- Protection of voter data using encrypted databases and secure access control.
- Improved system resilience using an intrusion detection mechanism to monitor unauthorized activities.
- Real-time authentication feedback for smooth voting operations.

1.2.3 PROBLEM DEFINITION

The problem addressed in this project is the absence of a robust and integrated voter verification system that can securely authenticate voters using digital IDs, biometrics, and intrusion detection while maintaining data integrity. The goal is to design and implement a Digitalized Voter-ID System with Fingerprint Authentication and Intrusion Detection that eliminates impersonation, enhances security, and ensures reliable verification in election environments.

1.3 OBJECTIVE

The primary objective of this project is to design and implement a secure, digitalized, and automated voter verification system that integrates QR-based voter identification, biometric fingerprint authentication, encrypted data management, and intrusion detection mechanisms to enhance the accuracy, transparency, and security of the electoral process.

1.3.1 SUB OBJECTIVES

1.To develop a QR-code-embedded Digital Voter-ID card that ensures fast, reliable, and tamper-resistant retrieval of voter information during initial verification.

2.To implement a fingerprint-based secondary authentication system using a biometric sensor to accurately validate voter identity and prevent

impersonation.

3.To design and maintain a secure, encrypted database architecture for storing voter demographic details, biometric templates, and enrollment records, thereby ensuring confidentiality, integrity, and protection against unauthorized modification.

4.To incorporate an intrusion detection mechanism for monitoring unauthorized system access, identifying anomalies, and notifying administrators of potential security breaches.

5.To provide voters and operators with instant authentication status updates through an LCD interface, ensuring transparency and seamless interaction during the verification process.

6.To implement encrypted communication protocols for secure data transfer between system modules, preventing data interception or manipulation during operation.

7.To develop a structured troubleshooting workflow for identifying, logging, and resolving authentication failures, sensor errors, database access issues, and system-level faults.

8.To perform comprehensive testing involving both successful and failed authentication cases, ensuring the system's accuracy, reliability, robustness, and fault tolerance under various operating conditions.

9.To design the system architecture to be scalable and adaptable, enabling future enhancements such as additional biometric modalities or multi-station network integration.

1.4 THRUST AREA

Digital Security and Encryption: The system emphasizes secure data handling through encryption techniques to protect sensitive voter information from tampering or unauthorized access.

Biometric Authentication Systems: The integration of fingerprint verification enhances identity accuracy and supports modern biometric security frameworks.

Embedded Systems and Real Time Implementation: The project utilizes microcontrollers, sensors, and peripheral devices, contributing to advancements in embedded systems for real-world applications.:

Cybersecurity and Intrusion Detection: By incorporating intrusion detection mechanisms, the project supports ongoing technological efforts toward securing digital infrastructures.

E-Governance and Digital Transformation: The system contributes to the modernization of government operations through digital voter management and automated authentication.

1.5 SOCIAL RELEVANCE.

Building Voter Confidence: Biometric authentication ensures that only genuine voters participate, increasing public trust in election outcomes.

Minimizing Manual Errors: Automation reduces human intervention, minimizing verification errors, delays, and potential bias.

Protecting Voter Data: Encrypted storage safeguards sensitive demographic and biometric information, ensuring high standards of data privacy and security.

Supporting Digital Governance: The system complements national digital initiatives by promoting secure, technology-driven voter management.

Improving Operational Efficiency: Quick and automated verification shortens queues and enhances the overall voting experience for citizens.

1.6 CONCLUSION

In this chapter the detailed view about the project objective, Problem statement and social Relevance were discussed.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

This chapter deals with the Literature survey for this project.

2.2 LITERATURE SURVEY

2.2.1 PROPOSED METHODOLOGY

Proposed Methodology presents the design and implementation of a Digitalized Voter-ID System with Fingerprint Authentication and Intrusion Detection, aimed at enhancing the security and reliability of voter verification in electoral processes. The proposed system employs a digital voter identification card embedded with a QR code, which serves as the initial verification element for fetching voter information from an encrypted database. An encrypted database is utilized to store voter profiles, verified enrollment records, and biometric fingerprint templates, ensuring data confidentiality, integrity and availability.

2.2.2 EXISTING METHODOLOGY

- 1 Amitesh Yadu, Prakash Chandrakar. “A Smart Voting System Combining Fingerprint and Facial Recognition for Enhanced Security.” work integrates dual biometric modalities fingerprint and facial recognition—to strengthen voter authentication. The system minimizes impersonation by requiring successful verification from both sensors before vote casting. It emphasizes real-time identity matching, secure database handling, and automated verification workflows. The authors highlight improved accuracy, reduced polling fraud, and enhanced reliability through multimodal biometrics.

- 2 Rosli Ismail, N. N. Zulkifli, M. M. Magiman, Mohd Saufi M. R. “Realtime Fingerprint based Voting System.”, a fingerprint-based voting platform designed for fast authentication and real-time verification. The system uses a fingerprint scanner integrated with a microcontroller and centralized database to validate voter identity instantly. The work focuses on preventing multiple voting attempts, minimizing human error, and improving election transparency using a secure, automated verification mechanism.
- 3 Madhuri Namballa, Tejasree Kaka, Mendu Vaishnavi, Duvvuri Sai Suma, K. Sriram. “Realtime Fingerprint based Voting System.” an electronic voting system that uses fingerprints as the primary authentication factor. The design includes fingerprint enrollment, encrypted template storage, and real-time matching during elections. It aims to eliminate fake voting, reduce administrative workload, and ensure accurate voter verification. The system architecture highlights ease of use, automation, and improved security over traditional voting methods.
- 4 P. J. Bharani, B. Gopinath. “Biometric Smart Card for Voter Identity with Secured Fingerprint Reader.”, a smart-card-based voting authentication system supported by a secure fingerprint reader. Voter identity is stored on an embedded smart card which must match a live fingerprint scan before enabling the voter to proceed. The approach strengthens identity verification, prevents card misuse, and supports encrypted communication between card, reader, and server. The authors argue that decentralized smart-card-based authentication enhances security.
- 5 Abhijeet Chavanke, Pritika Somase, Jayshri Avhad, Pranjali Salve, A. S. Dalvi. “An Electronic Voting System Using Fingerprint Authentication.”, an electronic voting platform where fingerprint identification ensures that voting is restricted to legitimate, registered

individuals. The system automates the verification process, maintains an encrypted voter database, and ensures each user can vote only once. The authors highlight improved operational efficiency, reduced fraud, and a user-friendly interface suitable for large-scale elections.

2.3 CONCLUSION:

In this chapter the review about the literature survey for the proposed project has been discussed.

CHAPTER 3

COMPONENTS DESCRIPTION AND SYSTEM INTERGRATION

3.1 INTRODUCTION

This chapter deals with the features and usage of each component which are used in this project along with the advantages.

3.2 MICROCONTROLLER -ESP32

The ESP32 development board is a powerful and versatile microcontroller used in our system as the central processing and communication unit. It handles sensor data collection, local processing, and secure wireless communication. With its dual-core processor, built-in Wi-Fi and Bluetooth, and extensive GPIO support, the ESP32 is well-suited for advanced IoT applications requiring speed, reliability, and real-time connectivity.

High-Speed Data Processing: The ESP32's dual-core 32-bit processor enables fast and efficient computation, allowing real-time processing of sensor data, authentication tasks, and system control operations.

Peripheral Expansion: With its ADCs, DACs, GPIOs, I²C, SPI, UART, and PWM support, the ESP32 interfaces easily with fingerprint sensors, cameras (OV2640), RFID modules, displays, and intrusion detection modules.

Real-Time Event Handling: The dual-core architecture allows one core to handle sensor tasks while the other manages connectivity, improving responsiveness and system stability.

3.2.1 ADVANTAGES OF USING ESP32

High Performance: The dual-core Tensilica processor offers robust computational power, supporting multitasking, encryption, image processing, and real-time security features.

Built-in Wi-Fi and Bluetooth: Enables seamless connectivity for IoT applications, peer-to-peer communication, and integration with mobile devices.

IoT-Optimized Architecture: Designed for scalability, real-time communication, and cloud integration, making it ideal for smart, automated, and security-focused systems.

Low Power Consumption: ESP32 includes multiple sleep modes and energy-saving features, making it suitable for battery-powered or portable devices requiring long-term operation.

Enhanced Security Features: Includes hardware encryption, secure boot, flash encryption, and WPA3 support—critical for applications involving authentication and intrusion detection.

Rich Peripheral Support: Supports multiple communication protocols (UART, SPI, I²C, CAN), camera modules, capacitive touch inputs, and analog/digital sensors.

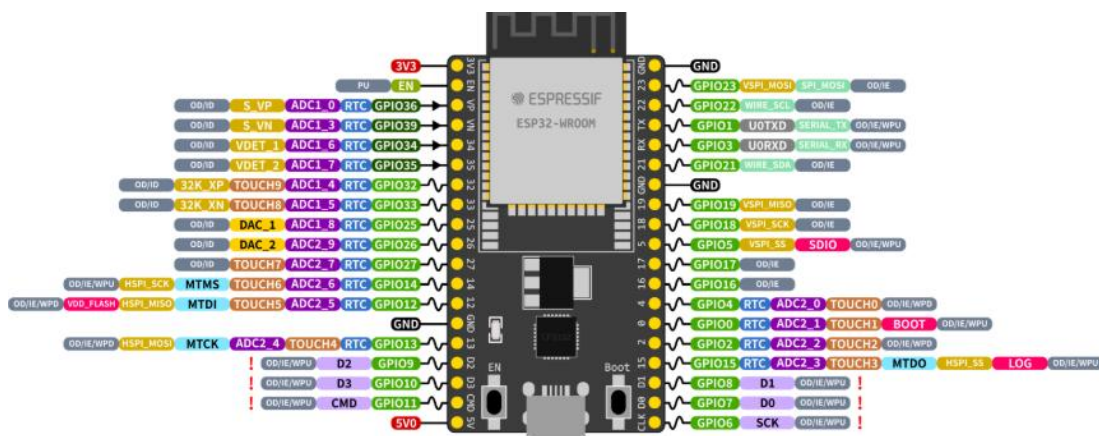


FIG 3.1. PINOUT DIAGRAM OF ESP32.

3.3 R307S FINGERPRINT SENSOR MODULE

The R307S is an enhanced version of the optical fingerprint sensor commonly used for biometric identification in secure embedded applications. It is designed to capture high-resolution fingerprint images and convert them into digital templates for authentication. The R307S provides improved performance, higher accuracy, and more reliable matching compared to earlier models, making it ideal for high-security verification systems.

In the Digitalized Voter-ID System with Fingerprint Authentication and Intrusion Detection, the R307s acts as the second stage of verification after QR code validation. Once a voter is marked as eligible, the R307S verifies their fingerprint against the stored biometric template to ensure that only authentic, registered users are granted access.



Fig 3.2 R307s Fingerprint Sensor Module

3.3.1 SENSOR SPECIFICATIONS

SPECIFICATION	DETAILS
Operating Voltage	3.6V – 6.0V
Operating Current	50 mA (typical), 120 mA (peak)
Interface	UART (TTL)
Baud Rate	9600 – 57600 bps (default: 57600)

SPECIFICATION	DETAILS
Fingerprint Capacity	1000 templates
Image Resolution	500 dpi
Image Array Size	256 × 360 pixels
Comparison Mode	1:1 (verification) and 1:N (identification)
FAR	< 0.001%
FRR	< 0.1%
Matching Speed	< 0.8 seconds
Security Level	Adjustable (1–5)

3.3.2 WORKING OF R307S

Fingerprint Image Capture: When a finger is placed on the sensor window, the optical module illuminates the finger surface and captures a high-quality grayscale fingerprint image using a built-in camera.

Feature Extraction: The sensor extracts unique minutiae points from the captured fingerprint image (ridge bifurcations, ridge endings, patterns). These features are then converted into a digital template for storage or matching.

Template Storage: The R307S includes internal flash memory where templates can be stored in indexed locations (0–255). Stored templates are used for future comparisons during authentication.

1:1 Verification Mode: The sensor compares a newly captured fingerprint with a specific stored template. This is useful when verifying a known user ID.

1:N Identification Mode: In this mode, the sensor searches through all stored templates to find a match. This is commonly used in voting authentication systems to identify unknown users.

3.3.3 ADVANTAGES OF THE R307S

Integrated Functionality: The module integrates image collection and an algorithm processing chip, handling all low-level tasks of fingerprint detection and verification internally. This eliminates the need for an external DSP chip algorithm and simplifies development.

Easy Integration and Secondary Development: It features both TTL UART and USB interfaces, allowing direct connection to a wide range of microcontrollers (3.3V and 5V, like Arduino or Raspberry Pi) and PCs with simple adapters.

High Resolution: Captures detailed fingerprint images up to 500 DPI.

Cost and Power Efficiency: The R307S is characterized by its low power consumption, low cost, and small size, making it a viable and efficient option for battery-powered or budget-conscious projects.

3.4 DISPLAY (16X2 LCD WITH I2C MODULE)

The 16×2 LCD is a simple alphanumeric display capable of showing 16 characters per line over two lines. When paired with an I2C module (PCF8574), the display becomes highly efficient, requiring only two communication lines (SDA and SCL). This significantly reduces wiring complexity and enhances reliability in embedded applications. In our project, the 16×2 LCD with I2C module is used to display real-time voter verification results such as QR scan status, fingerprint authentication result, and intrusion alerts, improving system transparency and user interaction.



FIG 3.3 Display (16x2 LCD with I2C module)

3.4.1 SPECIFICATION

SPECIFICATION	DETAILS
Display Type	Character LCD (16 characters × 2lines)
Controller	HD44780 (via PCF8574 I2C adapter)
Interface	UART (TTL) / USB (optional)
Operating Voltage	4.7V – 5.5V
I2C Voltage	3.3V – 5V compatible
Communication	I2C (SDA, SCL)
Character Format	0x20 to 0x27
Pin Count (Module)	Only 4 pins: GND, VCC, SDA, SCL

3.4.2 WORKING OF 16×2 LCD WITH I2C MODULE

I2C Communication Handling: The PCF8574 I2C expander converts I2C serial commands from the ESP32 into parallel signals required by the LCD.

Character and Command Processing

The LCD processes two types of data:

- Commands (cursor position, clear screen, backlight control)
- Character Data (ASCII characters displayed on the screen)

Display Update: Characters are displayed in a 5×8 dot matrix format. The ESP32 updates the screen dynamically to show:

- a) QR validation results
- b) Fingerprint match/mismatch
- c) Voter eligibility status
- d) Intrusion warnings

Backlight and Contrast Control: Backlight can be toggled programmatically, while contrast is set using the onboard potentiometer on the I2C module for optimal visibility.

3.4.3 ADVANTAGES OF THE 16×2 LCD

Reduced Wiring: Requires only two data lines (SDA & SCL), simplifying the hardware setup significantly.

Low Power Consumption: Suitable for continuous operation in embedded systems.

Easy Integration: Works seamlessly with ESP32, Arduino, and other microcontrollers using standard I2C protocols.

Clear and Real-Time Display: Ideal for showing authentication and system messages during voter verification.

Cost-Effective and Reliable: Economical for large-scale projects while offering stable and readable output.

3.5 REAL-TIME CLOCK MODULE – DS3231

The DS3231 is a highly accurate and temperature-compensated Real-Time Clock (RTC) module used for maintaining precise timekeeping in embedded systems. It integrates a 32 kHz crystal oscillator, temperature sensor, and battery backup to ensure continuous time tracking even when the main system power is disconnected.

In our project, the DS3231 plays a critical role in timestamping voter verification events, logging intrusion attempts, and maintaining secure, traceable records of every authentication and system activity.



FIG 3.4 Real-Time Clock module (DS3231)

3.5 .1 RTC- SPECIFIATIONS

SPECIFICATION	DETAILS
Operating Voltage	2.3V – 5.5V
Interface	I ² C (SDA, SCL)
Timekeeping Accuracy	± 2 ppm (approx. ± 1 minute per year)
Temperature Range	-40°C to $+85^{\circ}\text{C}$
Backup Battery	CR2032 (ensures timekeeping even without main power)
Oscillator	Temperature-compensated 32 kHz crystal (TCXO)
Registers	Stores seconds, minutes, hours, day, date, month, year
Alarm Functions	Two programmable alarm registers
Square-Wave Output	1Hz, 4kHz, 8kHz, 32kHz

3.5.2 WORKING OF DS3231 RTC MODULE

Timekeeping Operation: The built-in temperature-compensated oscillator maintains extremely stable time regardless of environmental changes. The

module tracks:

Seconds, Minutes, Hours, Day, Date, Month, Year (with leap year correction)

I²C Communication: The module communicates with the ESP32 using the I²C protocol.

a) ESP32 sends commands to read or set time.

b) DS3231 responds with binary-encoded time data.

This makes integration simple and efficient.

Battery Backup System: The DS3231 includes a CR2032 battery socket that allows the module to continue running even when system power is off.

3.5.3 ADVANTAGES OF THE DS3231

Highly Accurate Timekeeping: Maintains precision over long periods with minimal drift.

Battery-Backed Operation: Ensures uninterrupted time tracking during power failures.

Easy Interface: Simple I²C communication makes integration with ESP32 seamless.

Built-in Alarms: Supports event scheduling and time-based triggers.

Temperature Compensation: Automatically adjusts for temperature variations, improving reliability.

Ideal for Logging Systems: Suitable for applications requiring timestamped records, such as voter authentication logs and intrusion tracking.

3.6 MicroSD Card along with Module:

Memory modules are essential components in embedded systems for storing configuration parameters, biometric templates, logs, and system data. In our project, memory components such as **FRAM**, **MicroSD Card**, and **EEPROM** are used to ensure reliable storage of voter records, authentication logs, intrusion events, and system configuration.



FIG 3.5 Micro SD Card along with module

3.6.1 SPECIFICATIONS

MicroSD card provides reliable non-volatile storage for small datasets such as system settings, flags, or state indicators.

SPECIFICATION

RAM Size

Use Case

DETAILS

16GB

Storing configuration hash values

3.6.2 ROLES OF MicroSD Card

MicroSD stores each vote details immediately after a voter is successfully authenticated and allowed to vote. The votes are stored in the hashed form in the MicroSD by using AES encryption. The votes are stored as each blocks in the SD memory.

3.7 CONCLUSION:

In this chapter the detailed description about the components used along with its specifications and advantages, how they are used in this proposed project has been discussed.

CHAPTER 4

QR CODE–BASED VERIFICATION

4.1 INTRODUCTION

In this chapter we shall discuss about the proposed Digitalized Voter-ID System with Fingerprint Authentication and Intrusion Detection, the QR code serves as the first stage of verification during the voting process. Each registered voter receives a unique QR-based Digital Voter-ID, generated during Phase-1 of the system. This QR code contains securely encoded voter information such as:

1. Voter ID Number
2. Name
3. Enrollment Reference

Scanning this QR code at the voting center allows the system to quickly retrieve the corresponding voter record from the database before shifting to the second stage.

4.1.2 ROLE OF QR CODE IN THE PROJECT

The QR code plays a central role across all phases defined in the *First Review* document:

Phase-1: Registration & QR Generation:

1. Voter enters demographic details on the web page.
2. System validates the data and generates a unique QR code.
3. QR code is issued as the voter's Digital Voter-ID.
4. The QR code and voter profile are stored in Firebase.

Phase-2: Eligibility Verification at Voting Center

- 1.The QR code is scanned using a QR reader module or mobile scanning interface.
- 2.ESP32 retrieves voter information from Firebase using the encoded ID.
- 3.LCD Display:
Eligible (Valid)
Not Eligible (invalid / fake QR / unregistered voter)
- 4.If eligible, the system prompts for Fingerprint Authentication.

Phase-3: Intrusion Detection

1. Invalid / fake QR scans are logged as suspicious activity.
2. Continuous failed scans trigger alerts.
3. DS3231 timestamps these events for forensic analysis.

4.1.2 ADVANTAGES OF USING QR CODES

They enable fast and accurate validation by allowing instant data retrieval and automatic scanning, which minimizes manual errors and speeds up the verification process at polling centers. Crucially, QR codes provide high security through encrypted voter IDs, ensuring that even a visually recreated code will fail backend verification, which is further supported by an intrusion detection feature that flags fake or invalid scans as suspicious actions. Finally, they are easy to generate and store, being lightweight digital elements that can be conveniently shared with voters via various digital platforms.

4.1.3 IMPORTANCE IN VERIFICATION

1. The project uses a two-factor authentication mechanism:
QR Code Verification (Stage-1)
Fingerprint Authentication (Stage-2)
2. The QR code ensures:
3. Only registered individuals can proceed.
4. Voters cannot skip biometric verification.

5. No duplicate identities enter the system.
6. Intrusion attempts are detected early.
7. It provides a secure bridge between digital registration (web interface) and physical verification (hardware system).

ELECTORAL COMMISSION OF E&I



NAME: VISHWA.R
FATHERNAME: P.RAMASAMY
SEX: MALE
DOB: 08/09/2003

Fig 4.1 QR-Voter ID

4.2 CONCLUSION

In this chapter the voter-ID's importance and the implementation was discussed.

CHAPTER 5

WEB PAGE

5.1 INTRODUCTION

In our project, the web interface was developed to provide a user-friendly platform where voters can securely register, authenticate, and access their digital voter-ID. The web page acts as the entry point for Phase-1 of the system, enabling users to sign up with their demographic details, generate their QR-based digital voter-ID, and later log in for verification.

The development focused on both front-end and back-end programming to ensure smooth data handling, secure authentication and encryption, and real-time access to stored voter information.

5.2 FRONT-END DEVELOPMENT

The front-end of the web application was designed to provide a simple, intuitive, and visually appealing user experience. The interface includes:

- Clean and responsive login page where voters enter their Email and Password to access the system.
- Sign-Up page that collects essential voter details such as Name, Age, Address, Email, Photo, and Fingerprint registration request.
- Voter dashboard to allow users to view their registration status and assigned voter ID.

Front-end Technologies Used:

- HTML – for structure
- CSS – for styling and responsiveness
- JavaScript – for dynamic operations and form validation

5.3 BACK-END DEVELOPMENT

The back-end of the system is responsible for securely storing all voter records and managing authentication processes. In this project, Firebase was used as the back-end service due to its real-time database, high security standards, and ease of integration.

Back-end Features

Stores voter data including Name, Age, Address, Email, QR-ID, and biometric references.

Generates and stores unique QR-coded voter IDs during Phase-1.

Validates login attempts and prevents unauthorized access.

Sends voter records to the ESP32 during Phase-2 for QR verification.

Stores system logs and timestamped events for intrusion tracking.

The Firebase back-end ensures

Fast data retrieval

High-level encryption

Robust authentication

Real-time database synchronization

5.3.2 USER INTERFACE – LOGIN & SIGNUP

The login and signup pages form the core of user interaction.

Voter Login Page

The login page includes

1. Email input
2. Password input
3. Login button
4. Link to “New voter? Sign Up”
5. The background features a high-resolution Indian flag, reflecting the theme of digital voting and national identity.

During login

1. Inputs are validated before submission
2. Firebase checks user credentials

3. Successful login redirects voters to their dashboard
4. Invalid details trigger alert messages

Voter Sign-Up Page

1. The sign-up page allows voters to:
2. Enter personal details
3. Upload a profile image
4. Register for fingerprint authentication (handled in Phase-2)
5. Generate their unique QR-coded voter ID
6. Upon successful registration:
7. A Digital Voter ID is generated
8. The record is saved in Firebase
9. The user can proceed to biometric verification at the voting center



FIG 5.1. LOGIN PAGE OF e-Voting WEBSITE.



FIG 5.2. Signup PAGE OF e-Voting WEBSITE.

5.4 SPECIAL CONCESSION FOR PwD

In the proposed method according to election commission of India, Commission is committed for building an equal access framework for Persons with Disabilities (PwDs) supported by the fundamentals of responsiveness, respect and dignity to enhance elector confidence among them; and support initiatives for improved service offerings and enhancing their electoral participation.

Electors / Voters having one among the 21 disabilities as mentioned by the Rights of Persons with Disabilities Act 2016 are termed as voters with disabilities. As of now, PwD voters are enrolled in electoral roll in following four categories

1. Visual impairment
2. Speech & hearing disability
3. Locomotor disability
4. Other

A PwD voter can register in any of the above categories by submitting Form 6.

The concession from the fingerprint authentication will be given provided that the respective individual should hold the Medical certificate for the Illness.



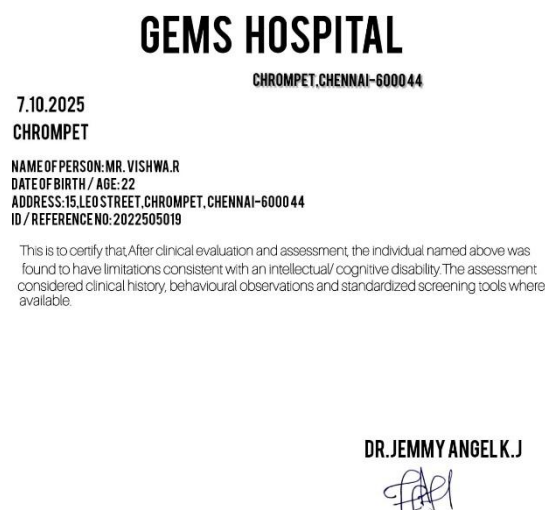
The image shows a digital form titled "Your Voter Information". At the top is a photo of a man. Below the photo is a table with the following details:

Name	Vishwa R
Father	P.Ramasamy
DOB	08/09/2003
Mobile	+91 6381655601
Sex	Male
Voter ID	44_2022505019
PWD Status	yes
Disability	Mental Illness

Below the table is a section labeled "Medical Certificate" which contains an image of a medical certificate from GEMS HOSPITAL. The certificate is dated 7.10.2025 and is for Mr. Vishwa R. At the bottom of the form is a blue "Logout" button.

Fig 5.3 Picture of Voter-ID with Medical Certificate.

Medical Certificate Attachment:



The image shows a medical certificate from GEMS HOSPITAL, CHROMPET, CHENNAI-600044. The certificate is dated 7.10.2025 and is for Mr. Vishwa R. The patient's details are: NAME OF PERSON: MR. VISHWA.R, DATE OF BIRTH / AGE: 22, ADDRESS: 15, LEQ STREET, CHROMPET, CHENNAI-600044, ID / REFERENCE NO: 2022505019. The certificate states: "This is to certify that, After clinical evaluation and assessment, the individual named above was found to have limitations consistent with an intellectual/ cognitive disability. The assessment considered clinical history, behavioural observations and standardized screening tools where available." The certificate is signed by DR. JEMMY ANGEL K. J.

Fig 5.4 Medical Certificate.

The medical certificate should get attached during the time of voter registration. The medical certificate should get signed by the authorized doctors by the government.

5.5 NOVELTY

1. A secure web-based voter identification system that generates unique digital Voter IDs and enables users to authenticate themselves, access their voter data, and understand the security layers safeguarding their vote.
2. Two Factor authentication enable enhanced protection in booth for voter-ID verification.
3. Intrusion Detection System blocks unauthorized voters and triggers alarm if any abnormalities are figured out.
4. Votes are stored in hashed format which is inaccessible for normal people and stored in secured environment (External Memory).
5. The integration of blockchain in vote enables vote tamper indication if any abnormalities in the blockchain is figured out.

5.6 CONCLUSION

In this chapter how the design of web page was made what are the technologies were used had been discussed. Also, the concession for PwD is elaborated.

CHAPTER 6

RESULT

6.1 INTRODUCTION

In the following chapter we will be seeing about the outcome of the proposed methodology along with the hardware implementation and Intrusion Detection.

6.2 VOTE STATUS

[illegible]

Fig 6.1 Hashed vote

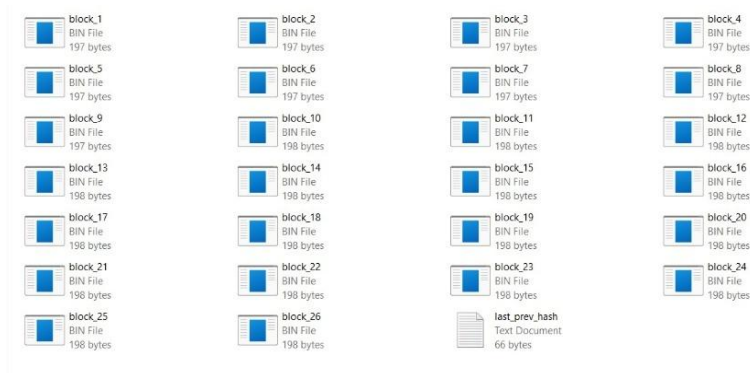


Fig 6.2 Stored blocks of vote in the MicroSD

In the fig 6.6 the status of the vote (i.e.) how the vote is stored in the block is shown. It is stored as 64 bytes of data. The fig 6.7 shows the blocks where the vote data are stored in the form of blocks. Each represents individual vote.

6.3 HARDWARE SETUP



Fig 6.3 Initial Step for Authentication



Fig 6.4 Eligibility check for the person

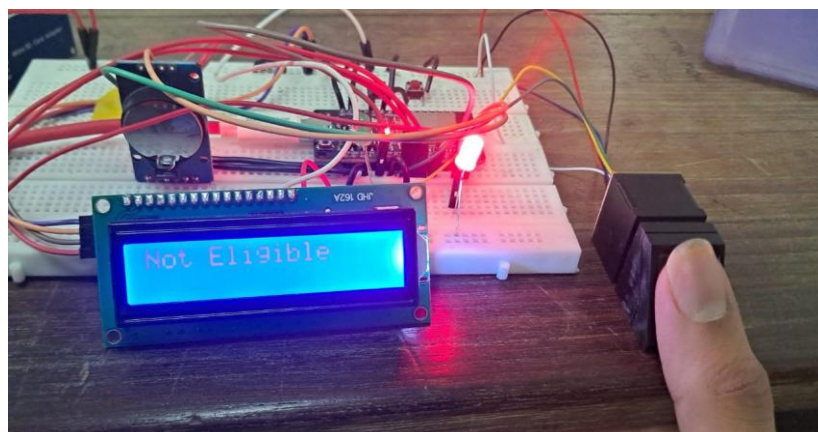


Fig 6.5 Not Eligible message is displayed

In the Hardware setup it is shown that the voter's name along with timestamp will be displayed. If the person is not eligible Red LED will glow and Buzzer will turn ON. If the person is eligible Green LED will turn ON. If same person Votes for second time "ALREADY VOTED -NAME" will be displayed along with that red LED and buzzer will be turned ON as an Indication of malpractice by the Intrusion Detection System.

6.4 VOTE RESULTS

<pre>WINNER = PARTY B Total A = 4 Total B = 5</pre>	<pre>Checking chain... Chain OK Decrypted vote: EVM-01 1 1 2025-11-23 20:16:08</pre>
---	--

Fig 6.6 Result declaration

Fig 6.7 Checking of Blockchain Status

By pressing the result button, the result will be declared for the particular booth. In the fig 6.5, It is shown that the "Chain is OK" which indicates that the result which are stored in the block are safe and no more tampered has occurred.

6.5 CONCLUSION

In this chapter the outcome of the proposed methodology and how the voter's authentication will happen in the real environment has been demonstrated and the storage of the casted vote along with its securing has been discussed.

CHAPTER 7

SUMMARY AND CONCLUSION

7.1 SUMMARY

The Digitalized Voter-ID System with Fingerprint Authentication and Intrusion Detection successfully demonstrates a modern, secure, and efficient approach to voter identification and verification. By integrating QR-based digital voter IDs with biometric fingerprint authentication, the system ensures that only legitimate and pre-registered voters proceed to the voting stage, thereby significantly reducing impersonation, duplication, and fraudulent entry attempts.

The first stage of verification, implemented through unique QR-coded voter IDs, enables fast and accurate identification of registered voters. This not only streamlines the verification process but also eliminates manual checking errors.

The second stage, fingerprint authentication using the R307S sensor, adds a robust biometric layer that ensures high accuracy and eliminates the possibility of unauthorized access. Additionally, the incorporation of intrusion detection mechanisms, such as logging failed scans, invalid QR attempts, and abnormal system activity, enhances the security and reliability of the overall system.

The project integrates both software (web portal, Firebase backend, QR generation) and hardware components (ESP32, fingerprint sensor, LCD, RTC, memory modules) into a unified system. This combination offers real-time processing, secure data handling, reliable time stamping, and user-friendly interaction. The use of a web-based registration platform developed using Visual Studio Code ensures efficient user onboarding and seamless QR-ID generation.

7.2 CONCLUSION

Overall, the system demonstrates how digital technologies can be leveraged to strengthen the integrity of election processes. The solution not only addresses common challenges found in traditional voting systems but also provides a foundation for future enhancements such as facial recognition, blockchain-based vote recording, and large-scale deployment.

The successful implementation of this project highlights the potential for secure, scalable, and technologically advanced voter identification systems capable of supporting transparent and fraud-free elections.

7.3 FUTURE SCOPE

The future scope for the proposed project is to implement this prototype for college level or school level election based on that enhancement can be made along with that third layer of enhancement in the security which includes IRIS scanner and Facial recognition shall be added.

REFERENCES

1. K. Kumar, Y. Singh. “A Review on Biometric Authentication Technologies and Their Applications in Secure Voting.” *International Journal of Information Security and Privacy* (2020): 150–158.
2. Madhuri Namballa, Tejasree Kaka, Mendu Vaishnavi, Duvvuri Sai Suma, K. Sriram. “Realtime Fingerprint-based Electronic Voting Machine.” *International Research Journal of Engineering and Technology (IRJET)* (2021): 204–210.
3. M. Hassan, F. Nur. “Anomaly-Based Intrusion Detection for Embedded IoT Platforms.” *Journal of Cybersecurity Research* (2021): 55–63
4. Rahul R., Priya M. “Design and Implementation of a Secure Voter Verification System Using Biometrics and QR Code Authentication.” *Proceedings of the International Conference on Smart Computing* (2020): 92–98.
5. J. George, T. Mathew. “Intrusion Detection Techniques for IoT-Based Security Systems.” *International Journal of Network Security* (2021): 67–75.
6. Abhijeet Chavanke, Pritika Somase, Jayshri Avhad, Pranjali Salve, A. S. Dalvi. “An Electronic Voting System Using Fingerprint Authentication.” *International Journal of Emerging Technologies* (2019): 33–40.
7. P. J. Bharani, B. Gopinath. “Biometric Smart Card for Voter Identity with Secured Fingerprint Reader.” *International Journal of Electronics and Communication Systems* (2020): 78–84.
8. Rosli Ismail, N. N. Zulkifli, M. M. Magiman, Mohd Saufi M. R. “Realtime Fingerprint-Based Voting System.” *IEEE Conference on Computer Applications* (2020): 112–118.

9. S. Das, P. Dutta. “Secured Electronic Voting System Based on Biometrics and Blockchain.” *IEEE Conference on Secure Computing* (2022): 88–95
10. Y. Zhou, L. Hu. “A Secure Mobile Voting System Using QR Code and Biometric Encryption.” *International Journal of Computer Applications* (2020): 50–56.